

Aishwarya Pawar

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Doctor of Information Technology student and Data Scientist with experience applying machine learning, statistical analysis, and generative AI to solve business problems. Brings hands-on experience translating complex data into practical, business-focused products and insights.

Education

Doctorate in Information Technology, Trine University, USA 07/2026 – Present
MSc., Data Science (Pure & Applied Math), Stevens Institute of Technology, NJ, USA, GPA: 3.96 09/2021 – 05/2023
Bachelor of Engineering, Electronics, Mumbai University, India 07/2013 – 08/2017
CFA Level 1 Candidate | Associate AI Engineer (Data Scientists and Developers) – *DataCamp*, 2025
Research Publications: [Time Series Analysis on Bitcoin and S&P 500](#) and [Data Analysis Using Statistical Methods](#)

Technical Skills

Programming & Tools: Python, SQL, PySpark, R, C++, Power BI, Tableau, Excel
Libraries & Frameworks: Pandas, NumPy, Scikit-learn, Pytorch, Hugging Face Transformers, FastAPI, statsmodels
Machine Learning & Statistics: Predictive Modeling, Classification, Regression, XGBoost, Random Forest, Clustering, A/B Testing, Statistical Inference, Model Explainability (SHAP, LIME), time series (Prophet, SARIMA, LSTM)
Generative AI & NLP: RAG, LLMs, BERT, LLaMA-2, Flan-T5, Hugging Face Transformers, LangChain, LangGraph, FAISS, LoRA, Prompt Engineering
Cloud & Data Pipelines: AWS, Databricks, Snowflake, Docker, Kubernetes, MLflow, CI/CD, Git/GitHub

Work Experience

Data Scientist, US Bank, Charlotte, NC, USA 09/2023 – 07/2026

- **Risk & Predictive Modeling:** Formulated and tested research hypotheses around default drivers, backtested them via PD/LGD models, and built credit score binning models for risk scoring, contributing to a 40% increase in funding volume
- **ML, DL & NLP Modeling:** Fine-tuned BERT to classify using extracted signals from unstructured notes; built ML and deep learning models to link borrower cash flow, repayment with macroeconomic data to identify early risk signals.
- **Generative AI:** Developed an internal RAG-based assistant using FAISS retrieval, LangChain LLM summarization, FastAPI, and AWS to improve access; designed a LangGraph based multi-agent proof of concept for credit workflows.
- **Customer Analytics & Model Monitoring:** Built XGBoost churn prediction and customer segmentation on Databricks for targeted retention; implemented MLflow to monitor performance and data drift; A/B testing showed 9% churn reduction.
- **Time-Series & Marketing Analytics:** Forecasted lead volume with Prophet/SARIMA with walk-forward validation and backtesting; analyzed channel, media mix, and uplift data to guide investment toward higher-converting channels.
- **Data Engineering & Reporting:** Architected a lakehouse-to-warehouse analytics pipeline using Python, SQL, and Databricks to publish governed Snowflake datasets; automated Power BI dashboards tracking KPIs for leadership.

Quantitative Financial Analyst Intern, Basis Point Global Solutions, Remote, USA 08/2022 – 01/2023

- **Quantitative Strategy:** Researched and back-tested equity selection strategies using sector and volume screens and ML models (LSTM, GARCH, Random Forest), applying VaR and Monte Carlo simulations to assess risk and evaluate returns.
- **Data Engineering:** Engineered AWS and Snowflake ETL pipelines to ingest Refinitiv data; built FASTAPI service to scrape news, summarized (Hugging Face Pegasus transformer), generate sentiment scores, to correlate with asset prices.

Data Scientist, TATA Elxsi, India 09/2018 – 07/2021

- **Predictive Modeling & Anomaly Detection:** Engineered predictive maintenance and anomaly detection ML pipelines in Python for high-velocity IoT sensor data, achieving 93.8% classification accuracy.
- **Model Deployment & Infrastructure:** Developed a performance-optimized C++ interface for real-time ML models and deployed containerized inference services using Docker and Flask.
- **Executive Analytics & BI:** Designed enterprise Tableau dashboards to translate complex ML prototype metrics and hardware telemetry into actionable insights for stakeholders.